comes to forming the Grothendieck semigroup or, by abuse of terminology, when we speak of two objects of $\mu\mathrm{VF}[k]$ being isomorphic.

Referee R167 [Typo; new] – Missing verb.
Original source line 1238. In the preceding paragraph, write "that is, *be* an isomorphism" after "should identify the two objects."

Definition 3.8 (RV-categories)**.** The objects of the category $\mathrm{RV}[k]$ are the pairs (U, f) with U a set in RV_∞ and $f : U \longrightarrow \mathrm{RV}^k$ a definable finite-to-one function. Given two such objects (U, f), (V, g), any definable bijection $F : U \longrightarrow V$ is a *morphism* of $\mathrm{RV}[k]$.

By Lemma 2.12, the two categories $\mathrm{VF}[0]$, $\mathrm{RV}[0]$ are actually equivalent, similarly for other such categories.

Notation 3.9. We emphasize that if (U, f) is an object of $\mathrm{RV}[k]$ then $f(U)$ is a subset of RV^k instead of RV^k_∞, while ∞ can occur in any coordinate of U. An object of $\mathrm{RV}[*]$ of the form (U, id) is often just written as U.

More generally, if $f : U \longrightarrow \mathrm{RV}^k_\infty$ is a definable finite-to-one function then (U, f) denote the obvious object of $\mathrm{RV}[\leq k]$. For example, the inclusion $\{\infty\} \longrightarrow \mathrm{RV}_\infty$ gives rise to an object of $\mathrm{RV}[0]$, the inclusion $\{(1, \infty)\} \longrightarrow \mathrm{RV}^2_\infty$ gives rise to an object of $\mathrm{RV}[1]$, and so on. Often f will be a coordinate projection (every object in $\mathrm{RV}[*]$ is isomorphic to an object of this form). In that case, $(U, \mathrm{pr}_{\leq k})$ is simply denoted by $U_{\leq k}$ and its class in $\mathbf{K}_+\,\mathrm{RV}[k]$ by $[U]_{\leq k}$, etc.

Referee R048 [Typo] – Agreement.
Original source line 1252. Change the verb "denote" to "denotes" in the sentence about (U, f).

Definition 3.10 (RES-categories)**.** The category $\mathrm{RES}[k]$ is the full subcategory of $\mathrm{RV}[k]$ such that $(U, f) \in \mathrm{RES}[k]$ if and only if $\mathrm{vrv}(U)$ is finite.

Notation 3.11. Every $\mathrm{RV}[k]$-morphism $F : (U, f) \longrightarrow (V, g)$ induces a definable finite-to-finite correspondence $F^\dagger \subseteq f(U) \times g(V)$, which may also be thought of as a finitary function $f(U) \longrightarrow \mathcal{P}(g(V))$. Recall the operator Jcb_Γ from Notation 2.8. For $u \in U$, we abbreviate $\mathrm{Jcb}_\Gamma\, F^\dagger(f(u), g \circ F(u))$ as $\mathrm{Jcb}_\Gamma\, F^\dagger(u)$

Referee R049 [Typo] – Missing punctuation.
Original source line 1260. Add a full stop after the abbreviation $\mathrm{Jcb}_\Gamma\, F^\dagger(u)$.

Definition 3.12 (RV- and RES-categories with Γ-volume forms)**.** An object of the category $\mu\mathrm{RV}[k]$ is a definable triple (U, f, ω), where (U, f) is an object of $\mathrm{RV}[k]$ and $\omega : U \longrightarrow \Gamma$ is a function, which is understood as a *definable Γ-volume form* on (U, f). A *morphism* between two such objects (U, f, ω), (V, g, σ) is an $\mathrm{RV}[k]$-morphism $F : (U, f) \longrightarrow (V, g)$ such that for every $u \in U$,

$$\omega(u) = (\sigma \circ F)(u) + \mathrm{Jcb}_\Gamma\, F^\dagger(u). \tag{3.2}$$

The category $\mu\mathrm{RES}[k]$ is the obvious full subcategory of $\mu\mathrm{RV}[k]$.

Definition 3.13 (Γ-categories)**.** The objects of the category $\Gamma[k]$ are the finite disjoint unions of definable subsets of Γ^k. A definable bijection $h : I \longrightarrow J$ between two such objects is a morphism of $\Gamma[k]$ if and only if there is a definable bijection $H : I^\sharp \longrightarrow J^\sharp$ satisfying $\mathrm{vrv} \circ H = h \circ \mathrm{vrv}$. For finite disjoint unions, the full preimages and this equation retain the disjoint-union tags.
C001 — proposed explicit full-preimage condition. This is the condition used in the checked Lean category. It makes the assignment $I \mapsto (I^\sharp, \mathrm{id})$ respect isomorphisms; scissors and products then give the lower horizontal semiring homomorphism. This does not establish injectivity. The

original wording did not explicitly require full preimages, and the weaker reading allowing bijections of arbitrary RV subsets does not suffice.

For example, over $\mathbb{S} = \mathbb{C}((t))$, with $\mathrm{val}(t) = 1$ and $a = \mathrm{rv}(t)$, squaring gives a definable bijection

$$\{u : u^5 = a\} \longrightarrow \{v : v^5 = a^2\}, \qquad u \longmapsto u^2, \quad v \longmapsto v^3/a \ \text{ inversely.}$$

It contracts to $\{1/5\} \longrightarrow \{2/5\}$, but the two full RV value fibres are not $\mathbb{S}$-definably bijective. Indeed, choose $r^5 = a$ only for the proof. A putative bijection F of the full fibres would give $g(z) = F(rz)/r^2$, a definable bijection of $\mathbb{C}^\times$. The induced residue-field structure is ACF of characteristic zero, so g and its inverse are generically rational and g is generically a Möbius transformation. The ramification automorphism $r \mapsto \zeta r$, for a primitive fifth root ζ, fixes the ground and the residue field, and implies $g(\zeta z) = \zeta^2 g(z)$. Thus this Möbius transformation conjugates multiplication by ζ to multiplication by ζ^2. It must permute their common fixed-point set $\{0, \infty\}$, and hence is cz or c/z. Neither satisfies that equation. One can work in the Puiseux algebraic closure of $\mathbb{C}((t))$; any putative defining formula transfers to this ACVF submodel. This counterexample concerns well-definedness under the weaker reading, not monicity under the proposed full-preimage condition.

The category $\Gamma^{\mathrm{fin}}[k]$ is the full subcategory of $\Gamma[k]$ such that $I \in \Gamma^{\mathrm{fin}}[k]$ if and only if I is finite.

Remark 3.14. If $\gamma \in \Gamma$ is definable then it is in the divisible hull of $\Gamma(\mathbb{S})$. This does not mean that the definable set $\gamma^\sharp \subseteq \mathrm{RV}$ contains a definable point unless $\gamma \in \Gamma(\mathbb{S})$.

Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \longrightarrow & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \longrightarrow & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where the two vertical arrows are monomorphisms.

C002 — proposed provisional removal of horizontal monicity. No counterexample to horizontal injectivity under the full-preimage definition is asserted. The missing implication is

$$I^\sharp \simeq_{\mathrm{RV}[k]} J^\sharp \quad \Longrightarrow \quad I \simeq_{\Gamma[k]} J.$$

It requires a same-ground fibre-respecting replacement for an arbitrary RV bijection, retaining all tags. Degree zero and singleton Gamma domains in every arity have been checked; the general case is unresolved. Lemma 2.29 of *Integration in algebraically closed valued fields with sections*, once its hypotheses and liftable translation constants are established, supplies piecewise integral-affine lifts in the forward direction. It does not give this reflection implication. The positive tensor presentation uses the horizontal homomorphisms and does not require their injectivity. This proposal withdraws an unjustified assertion pending a proof or a counterexample; it does not declare that assertion false.

The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U,f)],[I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 (Positive tensor presentation)**.** *Let T_+ and $T_{\mu,+}$ be the sources of Ψ and $\mu\Psi$, respectively. All definitions below use the original ground $\mathbb{S}$. For a* RES *object with underlying set X, put*

$$\Lambda_X = \{\delta \in \Gamma \cap \mathrm{dcl}^{\mathrm{eq}}(\mathbb{S}) : \exists\ \mathbb{S}\text{-definable } c : X \longrightarrow \delta^\sharp\}.$$

Let $\mathcal{C}$ be the semiring congruence on T_+ generated by

$$[\boldsymbol{X}]_r \otimes [U]_m \ \sim\ [\boldsymbol{X}]_r \otimes [U + \delta e_i]_m, \qquad \boldsymbol{X} \in \mathrm{RES}[r],\ U \in \Gamma[m],\ \delta \in \Lambda_X,$$

where $m \geq 1$ and i is a coordinate index. Let $\mathcal{C}_\mu$ be the semiring congruence on $T_{\mu,+}$ generated by

$$[\boldsymbol{X}]_r \otimes [(U,\sigma)]_m \ \sim\ [\boldsymbol{X}]_r \otimes [(U + \delta e_i, \sigma_\delta)]_m, \qquad \sigma_\delta(\eta) = \sigma(\eta - \delta e_i) - \delta,$$

with $\boldsymbol{X} = (X, f, \omega) \in \mu\mathrm{RES}[r]$ and the same admissibility condition $\delta \in \Lambda_X$. The homomorphisms Ψ and $\mu\Psi$ are surjective, with kernel congruences $\mathcal{C}$ and $\mathcal{C}_\mu$, respectively. They induce isomorphisms of graded semirings

$$T_+/\mathcal{C} \ \simeq\ \mathbf{K}_+\,\mathrm{RV}[*], \qquad T_{\mu,+}/\mathcal{C}_\mu \ \simeq\ \mathbf{K}_+\,\mu\mathrm{RV}[*].$$

Proof. The finite product normal form over the original ground partitions every $\mathrm{RV}[k]$ object into pieces isomorphic to $X \times U^\sharp$, where X has finite Gamma support and has a finite label of degree $k - m$, and $U \in \Gamma[m]$. For a measured object the combined weight on a RES factor has finite original-ground Gamma image. Partition by these values and use the shared finite measured Gamma factor in degree zero to transfer each constant weight to the Gamma factor. We may therefore normalize every RES combined weight to 0 and retain the whole weight as a definable